

ECON 0100 | Homework B3 | Equilibrium

| Solutions

Answers and work are shown in the red blocks.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P = 100 - \frac{1}{2}Q_d$$

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q$$

Prices are in galleons and quantity is in bottles. Use the space on the second page to graph the market.

Q1 | A Price Below Equilibrium

Suppose the price is 50 galleons.

a) What is the quantity demanded? _____

100 bottles.

Read each curve at the going price.

$$50 = 100 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 50 \rightarrow Q_d = 100$$

b) What is the quantity supplied? _____

60 bottles.

$$50 = 20 + \frac{1}{2}Q_s \rightarrow 30 = \frac{1}{2}Q_s \rightarrow Q_s = 60$$

c) Is this a shortage or an excess, and how large? _____

A shortage of 40 bottles.

Buyers want 100 and sellers offer 60: $Q_d > Q_s$, a shortage of $100 - 60 = 40$ bottles. Only 60 bottles change hands, the smaller of the two.

d) Which way will the price move? _____

Up.

Buyers who cannot find a bottle offer more, and sellers see they can charge more and still sell out, so the price rises toward 60, where the shortage closes.

Q2 | A Price Above Equilibrium

Suppose instead the price is 70 galleons.

a) What is the quantity demanded? _____

60 bottles.

$$70 = 100 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 30 \rightarrow Q_d = 60$$

b) What is the quantity supplied? _____

100 bottles.

$$70 = 20 + \frac{1}{2}Q_s \rightarrow 50 = \frac{1}{2}Q_s \rightarrow Q_s = 100$$

c) Is this a shortage or an excess, and how large? _____

An excess of 40 bottles.

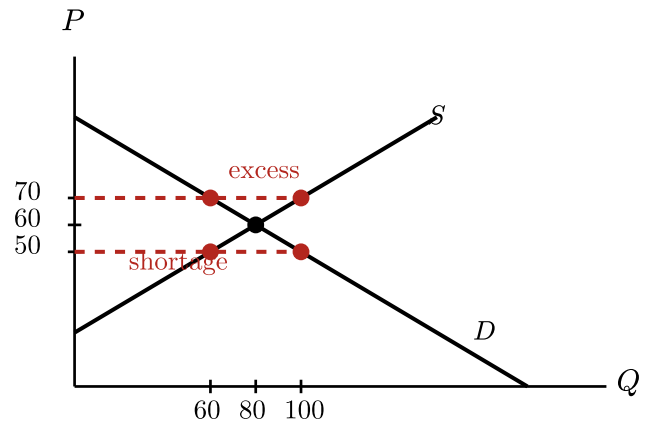
Sellers offer 100 and buyers take 60: $Q_s > Q_d$, an excess of $100 - 60 = 40$ bottles; again 60 are exchanged.

d) Which way will the price move? _____

Down.

Sellers stuck with unsold bottles cut the price, and buyers hold out for the lower offers, so the price falls toward 60.

The two prices are mirror images: 10 below equilibrium leaves 40 bottles short, 10 above leaves 40 unsold, and either way the market moves back to the equilibrium found in Q3.



Q3 | Equilibrium

a) Calculate equilibrium price: _____

60 galleons.

The prices of Q1 and Q2 both pushed the market toward one point. At equilibrium the same price and quantity sit on both curves, $Q_d = Q_s$, so set the two prices equal, solve for Q , and read the price off either curve:

$$100 - \frac{1}{2}Q = 20 + \frac{1}{2}Q \rightarrow 80 = Q \rightarrow Q^* = 80$$

$$P^* = 20 + \frac{1}{2} \cdot 80 = 60 \quad (= 100 - \frac{1}{2} \cdot 80)$$

b) Find equilibrium quantity: _____

80 bottles.

c) Calculate consumer surplus: _____

1600 galleons.

Consumer surplus is the triangle below demand and above P^* , out to Q^* :

$$CS = h \cdot b \cdot \frac{1}{2} = (100 - 60) \cdot 80 \cdot \frac{1}{2} = 1600$$

d) Calculate producer surplus: _____

1600 galleons.

Producer surplus is the triangle above supply and below P^* , out to Q^* :

$$PS = h \cdot b \cdot \frac{1}{2} = (60 - 20) \cdot 80 \cdot \frac{1}{2} = 1600$$

The two triangles are equal because the curves have the same slope, so P^* sits midway between the intercepts.

